

❖ **Easy level**

1. Even or Odd

Ask the user to enter a number and check whether it is even or odd using an if-else statement.

2. Largest of Three Numbers

Write a program that takes three numbers as input and prints the largest one using if-else.

3. Temperature Converter

Convert a given temperature from Celsius to Fahrenheit using the formula:

$$F = (C \times 9 / 5) + 32$$

4. Simple Discount Calculator

Ask the user to enter the price of an item. If the price is above \$100, apply a 10 % discount; otherwise, no discount applies. Print the final price.

5. Grading System

Take a student's score as input and print the corresponding grade:

- 90 - 100: A
- 80 - 89: B
- 70 - 79: C
- 60 - 69: D
- Below 60: F

6. Swap Two Numbers

Ask the user to enter two numbers and swap them.

7. Days to Weeks and Days Converter

Ask the user to enter a number of days and convert it into weeks and remaining days.

8. Electricity Bill Calculator

Ask the user for the number of units consumed and calculate the electricity bill based on these rules:

- Up to 100 units → \$0.5 per unit
- 101 - 300 units → \$0.75 per unit
- 301 + units → \$1 per unit

9. Simple Calculator

Write a program that takes two numbers and an operator (+, -, *, /) as input and performs the operation.

❖ **Medium level**

1. Basic ATM System

- Use a starting balance of \$1000.
- Allow users to Deposit, Withdraw, or Check Balance.
- Withdraw should check if funds are available.
- Display a menu with options:
 - o 1. Withdraw Money
 - o 2. Deposit Money
 - o 3. Check Balance
 - o 4. Exit

2. Factorial of a Number

- Ask the user for a number N and calculate its factorial(N!).

3. Sum of Even and Odd Numbers

- Ask the user to enter N.
- Find the sum of all even and odd numbers from 1 to N .

4. Guess Game with helper

Generate a random number(1 - 100) and ask the user to guess the number.

- If the guess is too high → "Try Lower!"
- If the guess is too low → "Try Higher!"
- If the guess is correct → "You Win!"

5. Print Pattern(For Loop)

- Ask the user for a number N and print a right - angled triangle
- Example: (for N = 5)

```
*
**
***
****
*****
```

6. Print a Pyramid Pattern(For Loop)

- Ask the user for N and print a pyramid
- Example: (for N = 4)

```
  *
 ***
*****
*****
```

7. Print a Diamond Pattern (For Loop)

- Ask for N, then print a diamond shape.

Example: (for N = 3)

```
*
***
*****
***
*
```

❖ **Hard level**

1. Arrays & Slices

1. Initialization & Output

- o Declare an array and a slice of 5 integers.
- o Assign values and print them using a loop.

2. Find Maximum & Minimum in a Slice

- o Ask the user to enter N numbers.
- o Find and display the maximum and minimum values.

3. Count Even & Odd Numbers

- o Ask the user to input N numbers into a slice.
- o Count and display how many numbers are even and odd.

4. Reverse a Slice

- o Ask the user for N numbers.
- o Reverse the inputs and print the result.

5. Search for a Number in a Slice

- o Input N numbers into a slice.
- o Ask for a number to search for and check if it exists.
- o If found, display its index.

6. Sorting a Slice

- o Ask the user for N numbers.
- o Sort them in ascending order.

- o Bonus: Sort them in descending order.

7. Remove Duplicates from a Slice

- o Ask the user to enter N numbers.

- o Remove duplicate values and print the unique numbers.

8. Second Largest Number

- o Input N numbers into a slice.

- o Find and display the second-largest number.

2. Strings

1. Count Vowels

- o Ask the user to enter a string.

- o Count and display the number of vowels in the string.

2. Reverse a String

- o Ask the user to enter a string.

- o Reverse the string and print the result.

3. Palindrome Checker

- o Ask the user to enter a string.

- o Check whether the string is a palindrome or not.

4. Count Character Frequency

- o Ask the user to enter a string.

- o Count how many times each character appears.

- o Display the result.

3. Maps

1. Student Grades Lookup

- o Create a map where the key is the student's name and the value is the student's grade.

- o Ask the user for a student's name and display the corresponding grade.

2. Word Frequency Counter

- o Ask the user to enter a sentence.
- o Count how many times each word appears.
- o Display the result.

3. Phone Book System

- o Create a map to store contact names and phone numbers.
- o Display a menu with options:
 - o 1. Add Contact
 - o 2. Search Contact
 - o 3. Delete Contact
 - o 4. Display Contacts
 - o 5. Exit

4. Inventory System

- o Create a map where the key is the product name and the value is the quantity.
- o Allow the user to:
 - o Add Product
 - o Update Quantity
 - o Search Product
 - o Display Inventory

4. Structs

1. Student Information

- o Create a Student struct containing:
 - o ID
 - o Name
 - o Age

- o Ask the user to enter the student's information and display it.

2. Multiple Students

- o Create a slice of Student.
- o Allow the user to enter information for multiple students.
- o Display all students.

3. Search Student by ID

- o Store multiple students in a slice.
- o Ask the user for a student ID.
- o Search and display the student's information.

4. Highest Grade Student

- o Create a Student struct containing:
 - o Name
 - o Grade
- o Store multiple students.
- o Find and display the student with the highest grade.

5. Employee Salary Calculator

- o Create an Employee struct containing:
 - o Name
 - o Monthly Salary
- o Calculate and display the employee's yearly salary.

